

TIMEBENCH: A Comprehensive Evaluation of Temporal Reasoning Abilities in Large Language Models

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Abstract

Grasping the concept of time is a fundamental facet of human cognition, indispensable for truly comprehending the intricacies of the world. Previous studies typically focus on specific aspects of time, lacking a comprehensive temporal reasoning benchmark. To address this, we propose TIMEBENCH, a comprehensive hierarchical temporal reasoning benchmark that covers a broad spectrum of temporal reasoning phenomena. TIMEBENCH provides a thorough evaluation for investigating the temporal reasoning capabilities of large language models. We conduct extensive experiments on GPT-4, LLaMA2, and other popular LLMs under various settings. Our experimental results indicate a significant performance gap between the state-of-the-art LLMs and humans, highlighting that there is still a considerable distance to cover in temporal reasoning. Besides, LLMs exhibit capability discrepancies across different reasoning categories. Furthermore, we thoroughly analyze the impact of multiple aspects on temporal reasoning and emphasize the associated challenges. We aspire for TIMEBENCH to serve as a comprehensive benchmark, fostering research in temporal reasoning¹.

1 Introduction

Time flies over us, but leaves its shadow behind.

Understanding time is a crucial part of human comprehension of the world. Envision the blossoming of flowers, and you'll associate it with the arrival of spring. The ponder within it encompasses the intricate interplay of world knowledge, causality, and event temporal relationships. Temporal reasoning, in contrast to reasoning of a singular nature, comes with inherent complexity, encompassing implicit arithmetic, logical implications, and world knowledge. It is a form of integrated reasoning built upon

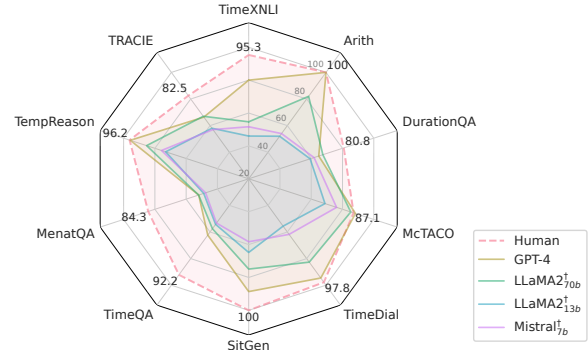


Figure 1: A brief overview of human and LLMs’ performance on TimeBench. Human scores are annotated.

foundational reasoning like mathematical and logical reasoning (Cobbe et al., 2021; Mishra et al., 2022; Yu et al., 2020). Recently, large language models (LLMs) have demonstrated remarkable performance in complex reasoning (Hendrycks et al., 2021; Srivastava et al., 2022; Brown et al., 2020; Chowdhery et al., 2023; OpenAI, 2023; Touvron et al., 2023), but their performance in temporal reasoning has not yet been extensively explored.

Recent research for temporal reasoning typically focuses only on a few aspects, such as temporal commonsense or temporal question answering (Zhou et al., 2019; Chen et al., 2021; Dhingra et al., 2022; Wang and Zhao, 2023). Due to the inherent complexity of temporal reasoning, it is challenging to accurately measure models’ temporal reasoning capabilities based on limited aspects.

To address this issue, we propose TIMEBENCH, a comprehensive and hierarchical temporal reasoning benchmark. Specifically, drawing inspiration from the human cognitive process of transitioning from abstraction and concreteness to integration (Barsalou et al., 2018), we categorize temporal reasoning into three levels: symbolic temporal reasoning, commonsense temporal reasoning, and event temporal reasoning. These levels respectively represent understanding abstract time expression,

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¹Code and data are available at: [GitHub](#)

grasping concrete world knowledge, and integrating and applying this knowledge in real-world scenarios. TIMEBENCH comprises 10 tasks with 16 sub-tasks, covering a broad spectrum of temporal reasoning phenomena. Besides, prior work typically features only a single task form, too simplistic to capture the model’s performance. In contrast, we incorporate four distinct task forms, offering a more realistic simulation of challenges.

To quantify the temporal reasoning capabilities of contemporary LLMs, we extensively assess widely-used LLMs, including proprietary models such as ChatGPT (Ouyang et al., 2022) and GPT-4 (OpenAI, 2023), as well as open-source like LLaMA2 (Touvron et al., 2023), Vicuna-1.5 (Chiang et al., 2023), Mistral (Jiang et al., 2023), Baichuan2 (Yang et al., 2023a), ChatGLM3 (Zeng et al., 2023) and FLAN-T5 (Chung et al., 2022). We conduct experiments under zero-shot and few-shot settings, combining commonly used reasoning techniques, chain-of-thought prompting (Kojima et al., 2022; Wei et al., 2022). The experimental results suggest that GPT-4 outperforms other models, showcasing strong temporal reasoning capabilities, as shown in Figure 1. Nevertheless, there is still a considerable gap between the strongest models and humans. On the contrary, open-source models show inferior performance in temporal reasoning, attributed to shortcomings in abstract time understanding, temporal relations modeling, and a lack of temporal commonsense. In addition, we also observe that chain-of-thought prompting does not yield a consistent improvement in performance. These findings indicate that there is still significant room for improvement in models’ temporal reasoning capabilities. Moreover, we have conducted a thorough analysis of the deficiencies and obstacles faced by models in temporal reasoning.

We aspire for temporal reasoning to garner increased attention within the research community. Our contributions can be summarized as follows:

- We introduce TIMEBENCH, a comprehensive and hierarchical benchmark to quantify the temporal reasoning abilities of LLMs.
- We conduct extensive experiments with several LLMs, revealing a significant gap between even SOTA LLM and humans, indicating substantial research opportunities in this field.
- By conducting a thorough analysis, we reveal the dilemmas that LLMs face in temporal reasoning and identify potential solutions.

2 TIMEBENCH Benchmark

2.1 Benchmark Design Principal

TIMEBENCH focuses on a comprehensive evaluation of the temporal reasoning capabilities of large language models in challenging and complex scenarios. To achieve this goal, we summarize the difficulties and challenges faced in temporal reasoning, categorize them into three levels, and integrate diverse task formats to better align with the intricate nature of temporal reasoning.

Just as the human cognitive process unfolds from foundational cognition and conceptual understanding to practical reasoning, we delineate temporal reasoning into three hierarchical levels. Specifically, TIMEBENCH categorizes temporal reasoning into symbolic, commonsense and event temporal reasoning, covering 10 datasets with a total of 16 subtasks. (1) Symbolic Temporal Reasoning focuses on the comprehension of fundamental abstract temporal expressions. (2) Temporal Commonsense Reasoning emphasizes the mastery of temporal principles, concepts and world knowledge. (3) Event Temporal Reasoning concentrates on modeling the temporal relationships between events and times within authentic scenarios.

2.2 Difficulties and Challenges

We delineate the essential competencies and the challenges that arise from a human cognitive standpoint in the realm of temporal reasoning, and language models confront similar challenges. We present the dataset statistics, task formats, and the associated challenges in Table 7.

Time Expression Understanding Time expressions (TimeX) denote words or phrases that convey information about time and represent the simplest and most basic units of expressing time, such as *in April 2000*, *after 2008*. Grasping time expressions is the most foundational step in understanding temporal elements within the textual modality.

Temporal Commonsense assesses the understanding of temporal world knowledge, including event order, event duration, typical time, event frequency and stationary, which is crucial for language models to comprehend daily scenarios.

Event-Time Relations assesses the model’s grounding capability to establish temporal relationships between events and their temporal context, thereby enabling models to grasp the progression

DATE ARITH
Q: What is the time 2 year and 4 month before Mar, 1755 A: Nov, 1752
TIMEX NLI
Premise: On 28th May 1967, I graduated. Hypothesis: Before 23rd October 1920, I graduated. A: Contradiction

Table 1: Examples of symbolic temporal reasoning

MCTACO
C: Ransome looks after her as well as for young Fern Simon , who has declared her love for him. Q: How often do Ransome and Fern talk? O: each century, <u>once a day</u> , once a century, <u>every night</u>
TIME DIAL
Dialog: ... Person1: Do you go to work by train every day Person2: Yes . I commute <MASK> a week by train... O: <u>five days</u> , 25 days, a minute, <u>six days</u>
SITUATEDGEN
Keywords: axis, one day, one month, Earth, Moon A: <u>Earth</u> rotates on its <u>axis</u> once in one day. It takes <u>one month</u> for the <u>Moon</u> to rotate on its <u>axis</u> .

Table 2: Examples of commonsense temporal reasoning.

and transformations of events as they dynamically evolve through time.

Event-Event Relations not only involve event-time grounding but also introduce multi-hop relative connections between events. Models with this capability can better handle temporal reasoning in complex scenarios involving multiple events.

Implicit Temporal Reasoning involves going beyond the surface of texts, engaging in deeper reasoning such as drawing upon temporal commonsense, identifying implicit temporal factors and discerning hidden temporal relationships among events. Implicit temporal reasoning is pivotal in complex real-world scenarios where events and time are intricately interwoven.

2.3 Symbolic Temporal Reasoning

To evaluate the language model’s comprehension of abstract time expressions, we utilize two symbolic reasoning tasks stripped of semantic content: date arithmetic and time expression inference. Table 1 shows examples of symbolic temporal reasoning.

TIMEQA
C: ... He worked in Utrecht for the firm of P Smits & de Wolf from 1864 to 1867 and then returned to ... Q: Where did Ludwig Mond work between Mar 1866 and Sep 1866? A: Utrecht
MENATQA
C: ... After the French evacuated Egypt in 1801, Hurshid Pasha was named governor of Egypt in 1804. Muhammad Ali had himself named governor of Egypt in May 1805 ... Q: Which position did Hurshid Pasha hold from 1804 to 1806, if Hurshid Pasha tepped down as the governor of Egypt in 1808? A: governor of Egypt
TEMPREASON
C: ... Peter Corke works for Queensland University of Technology from Jan, 2010 to Dec, 2022. Peter Corke works for Commonwealth Scientific from Jan, 1984 to Jan, 2009. ... Q: Which employer did Peter Corke work for before Queensland University of Technology? A: Commonwealth Scientific

Table 3: Examples of event temporal reasoning.

Date Arithmetic (Tan et al., 2023) assesses the model’s grasp of abstract date calculation. When provided with a date, the model needs to accurately calculate the date a certain amount of time before or after the given date. The smallest unit is one day.

TimeX NLI (Thukral et al., 2021) focuses on the logical entailment relationships among abstract TimeX, including three aspects: order (s1), duration (s2), and duration with unit conversion (s3).

2.4 Commonsense Temporal Reasoning

We measure the model’s mastery of temporal commonsense and world knowledge, along with its capacity for reasoning based on these insights. Table 2 presents examples of temporal commonsense reasoning in QA and generation forms.

MCTACO (Zhou et al., 2019) evaluates diverse commonsense knowledge from different aspects of events, including duration, frequency, order, stationary and typical event time.

DurationQA (Virgo et al., 2022) focuses specifically on temporal commonsense reasoning in the spectrum of event duration.

TimeDial (Qin et al., 2021) considers temporal commonsense reasoning in dialogue scenarios and involves various aspects of commonsense associated with duration, order, and world knowledge.

SituatedGen (Zhang and Wan, 2023) considers generative commonsense reasoning in a constrained text generation scenario. Given a set of contrasting keywords, the model needs to choose appropriate keywords for each sentence and generate a pair of contrasting sentences that satisfy temporal commonsense.

2.5 Event Temporal Reasoning

Event temporal reasoning assesses the model’s understanding of relationships between events and time in real-world scenarios, as well as its ability to reasoning under certain temporal or event constraints. Examples are shown in Table 3.

TimeQA (Chen et al., 2021) requires the model to answer time-sensitive questions based on context containing numerous time-involved facts. It is categorized into explicit reasoning and implicit reasoning based on time indicators (before, in, etc.).

MenatQA (Wei et al., 2023) introduces time-sensitive factors to elicit implicit temporal reasoning, including time scope change, disruption of facts, and counterfactual questions, which provides a more in-depth assessment of implicit reasoning ability on event-time relations.

TempReason (Tan et al., 2023) removes irrelevant context and focuses on implicit temporal reasoning within structured facts, investigating the model’s capability boundaries. It involves event-time reasoning and event-event reasoning.

TRACIE (Zhou et al., 2021) evaluates the model’s comprehension of temporal order between implicit events. The model needs to identify events implied in the context and then determine their chronological order.

2.6 Task Formats and Evaluation Metrics

TIMEBENCH is a multispectral benchmark encompassing four task types: free-form reading comprehension, natural language inference, constrained text generation, and multi-select questions. For detailed task types and their corresponding evaluation metrics, please refer to Appendix A.3 and A.4.

3 Methodology

We perform evaluations using the prompt-based approach, including standard prompting and chain-of-thought prompting. Experiments are conducted under both zero-shot and few-shot settings.

Standard Prompting We formulate specific instructions for each task. In the zero-shot setting, models follow the instructions to answer questions. In the few-shot setting, models are provided with several question-answer pairs as demonstrations and emulate those instances to answer questions.

$$\text{prompt}_{\text{zs}}^{\text{sp}} = \{\text{INST}\}\{\text{Q}\} \quad (1)$$

$$\text{prompt}_{\text{fs}}^{\text{sp}} = \{\text{INST}\}\{\text{Q}_1\}\{\text{A}_1\}..\{\text{Q}\} \quad (2)$$

Chain-of-Thought Prompting The instructions of CoT are the same as standard prompting. In the zero-shot setting, following Zeroshot CoT (Kojima et al., 2022), we add a reasoning trigger *Let’s think step by step* after questions to perform chain-of-thought reasoning. In the few-shot setting, we manually annotate CoT demonstrations for each task to guide the step-by-step reasoning. Prompts can be found in Appendix B.3.

$$\text{prompt}_{\text{zs}}^{\text{cot}} = \{\text{INST}\}\{\text{Q}\}\{\text{TRIG}\} \quad (3)$$

$$\text{prompt}_{\text{fs}}^{\text{cot}} = \{\text{INST}\}\{\text{Q}_1\}\{\text{R}_1\}\{\text{A}_1\}..\{\text{Q}\} \quad (4)$$

4 Experimental Setup

4.1 Models

We evaluate several popular LLMs, including both open-source and proprietary models, with parameter sizes ranging from 6B to 70B.² The complete list of models can be found in Appendix B.1.

4.2 Implementation Details

We access proprietary models through Azure API 0613 version. For open-source models, we deploy them locally through FastAPI. We set the temperature to 0.0 for greedy decoding in all experiments. To improve answer extraction accuracy, we prompt models with trigger *Therefore, the answer is* before model outputs to deduce final answers.

5 Experimental Results

5.1 Few-shot Results

Table 4 presents the experimental results under few-shot settings. GPT-4 achieves the best performance across three categories, while LLaMA2_{70b} and GPT-3.5 rank in the second tier. However, there remains a substantial gap of 19.4% between the most powerful LLM and humans.

In symbolic temporal reasoning tasks, GPT-4 demonstrates exceptional performance. However,

² Since OpenAI has never disclosed the scale of ChatGPT series, 6B to 70B here refers to ChatGLM3_{6B} to LLaMA2_{70B}.

Method	Symbolic				Commonsense				Event Temporal							Overall				
	TimeXNLI			Arith	DQA	McT.	TiD.	SitGen	TimeQA		MenatQA		TempR		TRACIE	Sym.	Comm.	Event	Avg.	
	s1	s2	s3						Exp.	Imp.	Scor.	Ord.	Ctf.	L2						L3
Human	98.0	96.0	92.0	100.0	80.8	87.1	97.8	100.0	93.3	91.1	85.6	87.3	79.9	97.1	95.3	82.5	96.5	91.4	89.0	91.5
GPT-4	85.3	73.3	53.3	100.0	64.8	88.3	94.6	88.6	73.7	51.0	72.4	54.8	28.7	92.4	95.9	62.8	78.0	84.1	66.5	73.7
+ FS CoT	92.0	84.0	64.0	100.0	55.1	72.3	93.4	-	66.9	52.8	65.3	52.6	25.9	96.9	94.6	66.4	85.0	73.6	65.2	72.1
GPT-3.5	52.0	68.4	31.6	63.6	67.7	71.2	76.4	79.1	66.1	48.4	43.2	51.6	17.9	84.7	78.0	55.0	53.9	73.6	55.6	59.7
+ FS CoT	51.6	71.8	36.6	84.4	41.2	38.1	71.1	-	68.0	47.0	42.5	41.7	37.8	89.9	76.6	50.2	61.1	50.1	56.7	56.6
LLaMa2 _{70b}	55.0	61.0	37.0	82.0	67.4	85.3	82.7	74.9	66.7	48.3	61.4	42.5	33.8	85.2	85.4	61.0	58.8	77.6	60.5	64.4
+ FS CoT	52.0	73.0	39.0	79.5	62.3	79.1	61.1	-	64.3	43.0	57.7	45.2	53.1	87.5	81.6	67.0	60.9	67.5	62.4	63.0
LLaMa2 _{13b}	50.0	54.0	30.0	29.5	53.3	66.0	55.6	64.8	59.3	48.6	49.6	43.4	37.5	78.7	62.7	58.0	40.9	59.9	54.7	52.6
+ FS CoT	40.0	61.0	37.0	52.0	59.3	68.8	40.8	-	59.4	49.1	58.4	43.8	44.1	78.0	68.2	58.0	47.5	56.3	57.4	54.5
LLaMa2 _{7b}	26.0	50.0	30.0	20.0	54.5	59.6	45.2	62.4	54.4	45.3	49.8	41.9	35.8	64.0	53.3	49.0	31.5	55.4	49.2	46.3
+ FS CoT	37.0	52.0	36.0	25.5	56.9	67.0	41.9	-	45.6	36.1	50.9	38.0	57.3	59.7	57.7	50.0	37.6	55.3	49.4	47.4
Baichuan2 _{13b}	38.0	48.0	33.0	42.5	54.8	73.0	45.7	64.9	59.4	54.2	52.7	38.0	21.4	77.3	63.5	54.0	40.4	59.6	52.6	51.3
+ FS CoT	50.0	56.0	34.0	47.0	62.0	69.3	43.8	-	58.2	49.6	49.8	40.1	45.6	81.3	65.6	60.0	46.8	58.4	56.3	54.2
Baichuan2 _{7b}	27.0	66.0	41.0	32.5	59.8	69.4	34.3	59.8	53.8	50.2	49.6	38.5	22.9	65.9	51.0	55.0	41.6	55.8	48.4	48.5
+ FS CoT	30.0	56.0	34.0	34.0	57.0	69.5	44.5	-	51.2	40.7	46.4	32.6	46.3	61.5	64.1	53.0	38.5	57.0	49.5	48.1
Mistral _{7b}	48.0	53.0	38.0	41.0	61.8	76.2	61.8	58.3	55.9	45.3	49.4	47.8	45.5	76.7	74.8	53.0	45.0	64.5	56.1	55.4
+ FS CoT	57.0	63.0	35.0	54.0	61.8	45.7	57.3	-	60.4	46.2	57.2	47.9	33.2	65.9	67.9	57.0	52.3	54.9	54.5	54.0
ChatGLM3 _{6b}	48.0	70.0	32.0	35.0	51.8	62.6	55.0	61.6	57.2	26.3	35.4	41.5	22.5	76.4	55.9	58.0	46.3	57.8	46.7	49.3
+ FS CoT	47.0	68.0	32.0	46.0	53.9	64.3	56.5	-	52.5	24.5	35.0	40.2	22.5	79.4	60.3	54.0	48.3	58.2	46.1	49.1

Table 4: Experimental results under **few-shot** settings (standard prompting by default). [†] denotes the base model without alignment. Global top-3 results are **bold**. Figure 8 provides a horizontal comparison of the performance of all models. Full results in Appendix B.2.

other models exhibit a significant decline in comparison to GPT-4. In commonsense temporal reasoning tasks, GPT4 lags behind humans by only 8.0%, indicating its powerful internal knowledge reservoir. With the model scale shrinking, its knowledge reservoir also decreases gradually, leading to a decline in performance. Notably, there is a significant gap of 25.2% between LLMs and humans in event temporal reasoning, which suggests that LLMs encounter major challenges in modeling intricate event-time relationships.

5.2 Zero-shot Results

Experimental results of alignment models under zero-shot settings are shown in Table 5. In zero-shot settings, GPT-4 and GPT-3.5 rank first and second, respectively, and they significantly outperform all open-source models by a large margin. It is noteworthy that open-source models exhibit a larger performance decline compared to proprietary models when transitioning from few-shot to zero-shot scenarios. GPT, Baichuan2 and LLaMA2 suffer drops of 5.6%, 14.6% and 27.2%, respectively. We attribute this performance decline to the quality of alignment. Restricted by their limited instruction-following capability, open-source models struggle to fully unleash their performance

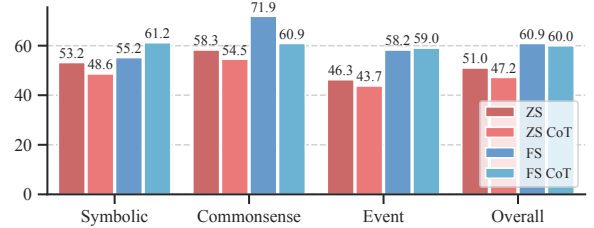


Figure 2: Performance gap with and without CoT prompting. The results are averaged from GPT-4, GPT-3.5, Baichuan2_{13b}, LLaMA2_{70b} and Mistral_{7b}.

solely through instructions. Therefore, few-shot prompting is a better approach for stimulating their temporal reasoning abilities.

5.3 Chain-of-Thought in Temporal Reasoning

Previous research has found that chain-of-thought prompting can enhance the model’s reasoning ability (Wei et al., 2022; Kojima et al., 2022). We aim to explore the following questions: *Does CoT prompting bring consistent improvement in temporal reasoning?* Due to the diversity of temporal reasoning, the above question has not yet been definitively answered. To investigate this, we select several popular LLMs and analyze their performance affected by chain-of-thought prompting.

Method	Symbolic				Commonsense				Event Temporal							Overall				
	TimeXNLI			Arith	DQA	McT.	TiD.	SitGen	TimeQA		MenatQA			TempR		TRACIE	Sym.	Comm.	Event	Avg.
	s1	s2	s3						Exp.	Imp.	Sco.	Ord.	Ctf.	L2	L3					
Human	98.0	96.0	92.0	100.0	80.8	87.1	97.8	100.0	93.3	91.1	85.6	87.3	79.9	97.1	95.3	82.5	96.5	91.4	89.0	91.5
GPT-4	78.6	76.0	50.7	98.0	59.2	80.0	91.1	59.3	60.6	46.5	57.0	57.0	23.1	95.3	95.0	64.8	75.8	72.4	62.4	68.3
+ CoT	80.0	76.0	60.0	92.0	58.1	82.6	89.3	-	61.3	41.2	54.6	59.6	22.6	97.0	94.5	58.0	77.0	76.7	61.1	68.5
GPT-3.5	45.4	67.6	31.2	97.0	50.5	68.6	69.1	62.3	70.8	35.4	40.9	43.9	22.9	81.2	73.8	57.4	60.3	62.6	53.3	57.4
+ CoT	33.6	64.8	33.6	71.0	23.2	45.1	67.0	-	64.4	35.1	39.7	42.9	26.3	57.6	68.1	52.0	50.8	45.1	48.3	48.3
LLaMA2 _{70b}	44.0	47.0	32.0	78.5	59.2	68.9	57.0	25.0	40.8	40.6	18.9	16.6	12.0	63.5	54.5	48.0	50.4	52.5	36.8	44.1
+ CoT	30.0	66.0	28.0	53.5	57.3	67.1	58.6	-	31.4	19.5	12.2	12.7	20.8	37.5	40.5	51.0	44.4	61.0	28.2	39.1
LLaMA2 _{13b}	30.0	49.0	34.0	22.5	38.5	40.6	35.4	57.9	61.9	30.5	46.1	36.1	26.9	53.1	69.4	49.0	33.9	43.1	46.6	42.6
+ CoT	36.0	50.0	38.0	6.0	39.2	51.7	36.9	-	58.7	38.9	40.9	32.5	33.6	58.0	68.4	47.0	32.5	42.6	47.3	42.4
LLaMA2 _{7b}	39.0	53.0	30.0	13.0	39.3	41.0	6.3	24.5	49.0	29.0	26.8	21.1	16.0	63.9	47.9	49.0	33.8	27.8	37.8	34.3
+ CoT	44.0	50.0	33.0	5.0	35.0	40.0	1.7	-	49.9	31.6	31.4	24.5	17.8	56.9	48.1	46.0	33.0	25.6	38.3	34.3
Baichuan2 _{13b}	41.0	61.0	37.0	12.5	52.0	63.4	57.7	52.2	55.4	34.6	48.8	44.3	39.5	57.4	61.4	49.0	37.9	56.3	48.8	48.0
+ CoT	40.0	57.0	31.0	10.0	44.6	61.9	58.1	-	41.5	40.9	52.0	38.5	43.2	62.8	64.3	55.0	34.5	54.9	49.8	46.7
Baichuan2 _{7b}	35.0	50.0	37.0	4.5	47.9	55.3	54.3	42.0	41.5	34.7	35.2	31.2	20.4	43.4	47.7	55.0	31.6	49.9	38.6	39.7
+ CoT	38.0	43.0	32.0	1.0	37.9	58.0	44.2	-	53.5	38.8	39.9	33.2	29.3	41.2	47.2	54.0	28.5	46.7	42.1	39.4
Vicuna1.5 _{13b}	35.0	50.0	36.0	15.0	39.2	59.1	34.2	51.8	60.4	37.0	46.8	37.4	23.2	42.1	43.6	46.0	34.0	46.1	42.1	41.1
+ CoT	42.0	51.0	37.0	3.0	29.8	50.0	33.7	-	56.9	36.4	38.2	37.7	20.4	49.0	49.1	51.0	33.3	37.8	42.3	39.0
Vicuna1.5 _{7b}	37.0	58.0	43.0	5.0	40.4	52.5	32.0	47.8	47.1	18.5	35.7	25.7	17.3	33.0	46.8	54.0	35.8	43.2	34.8	37.1
+ CoT	36.0	50.0	36.0	1.5	39.4	49.2	36.2	-	40.9	24.6	26.2	28.5	25.0	27.7	40.3	54.0	30.9	41.6	33.4	34.4
FLAN _{T5} _{11b}	53.0	63.0	43.0	0.0	52.0	65.0	47.7	49.5	61.7	26.8	33.6	52.2	21.8	87.9	83.9	64.0	39.8	53.6	54.0	50.3
+ CoT	56.0	66.0	45.0	0.0	49.7	63.4	42.7	-	64.4	28.2	41.6	50.2	30.6	79.5	68.9	55.0	41.8	51.9	52.3	49.4
Mistral _{7b}	47.0	50.0	43.0	26.5	49.8	58.8	23.2	58.3	28.2	21.4	24.3	22.3	21.7	39.6	31.6	51.0	41.6	47.5	30.0	37.3
+ CoT	38.0	56.0	35.0	16.5	36.6	49.3	19.3	-	31.3	22.4	21.1	24.9	25.6	34.0	31.2	61.0	36.4	35.1	31.4	33.5
ChatGLM3 _{6b}	38.0	50.0	34.0	2.0	34.1	43.6	56.7	38.9	41.2	31.7	33.8	26.0	32.2	57.0	54.0	50.0	31.0	43.3	40.7	39.0
+ CoT	27.0	49.0	37.0	0.0	24.8	37.1	44.8	-	41.7	25.4	34.6	28.1	41.2	44.5	52.0	48.0	28.3	35.6	39.4	35.7

Table 5: Experimental results under **zero-shot** settings (standart prompting by default). All models are alignment models (-chat or -instruct). Global top-3 results are **bold**.

Chain-of-thought reasoning is not consistently effective. As illustrated in Figure 2, introducing zero-shot CoT prompting results in consistent declines, with an overall decrease of 7.4%. In the few-shot scenario, CoT prompting also fails to yield consistent improvements, varying depending on the task. There is a 10.8% improvement in symbolic reasoning, while a significant decline of 15.2% in commonsense reasoning. In event temporal reasoning, there is a slight improvement of 1.3%. Next, we will conduct a more detailed analysis of the impact of CoT on specific tasks.

Impact of CoT prompting across tasks. In order to explore the impact of CoT on various tasks thoroughly, we delve into the performance changes of each model across specific tasks within each category, as illustrated in Figure 3. In the zero-shot setting, open-source models achieve a slight improvement in event temporal reasoning with chain-of-thought prompting, while in other cases, they face performance degradation. While in the few-shot setting, almost all models exhibit significant improvement in symbolic temporal reasoning,

with a concurrent prevalent decline in commonsense temporal reasoning. We attribute this to the knowledge sensitivity inherent in commonsense reasoning, where step-by-step reasoning cannot compensate for the lack of knowledge. In event temporal reasoning, improvements mainly stem from datasets involving implicit multi-step reasoning (MenatQA and TempReason), indicating that CoT is more effective for multi-hop questions. In summary, zero-shot CoT consistently has a negative impact on temporal reasoning. While in few-shot scenario, CoT has a positive impact on symbolic and complex tasks, while negatively affecting knowledge-sensitive tasks.

6 Analysis and Discussion

6.1 Scaling Effect of Model Size

We investigate how the scale of models affects temporal reasoning capabilities. The trend is illustrated in Figure 4. As the model scale increases, there is a notable improvement in performance. When the parameter size expands from 7B to 13B, LLaMA2 and Baichuan2 show improvements of

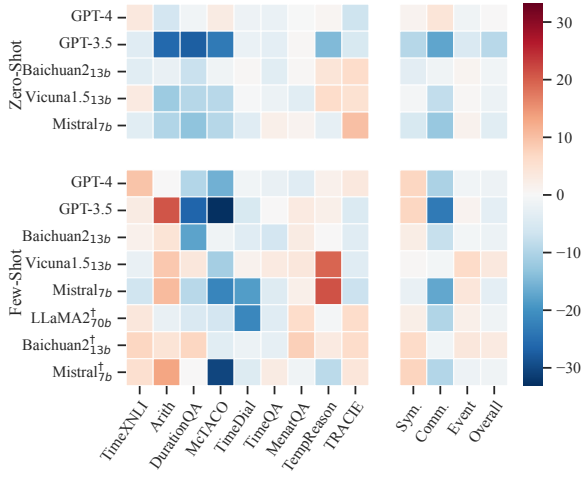


Figure 3: Δ Score between the chain-of-thought prompting and direct I-O prompting. **Top:** zero-shot setting, **Bottom:** few-shot setting, **Left:** variation in each task, **Right:** averaged variation in the symbolic, commonsense, event, and overall tasks.

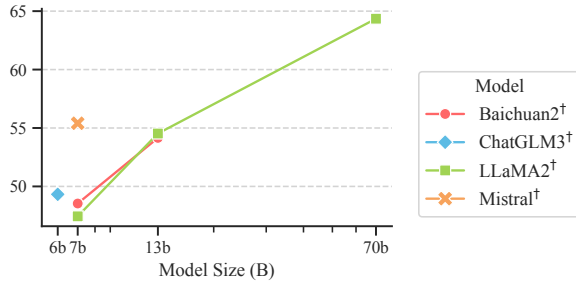


Figure 4: Scaling effect of model size and overall temporal reasoning performance. The x-axis (model size) is shown in the log scale. Results show a log-linearity between parameter size and performance.

13.0% and 10.5%, respectively. Furthermore, when LLaMA scales up to 70B, the trend of performance improvement continues without stopping. The overall improvement follows a log-linear relationship with scale. There are no significant performance differences among LLaMA2, Baichuan2, and ChatGLM3 under similar parameter specifications, while Mistral demonstrates impressive prowess, outperforming all other 13B models with nearly half the number of parameters.

6.2 Challenges in Temporal Reasoning

LLMs underperform in (multi-hop) symbolic reasoning Except for GPT-4, the performance of all other models in symbolic temporal reasoning is unsatisfactory. A noticeable decrease is observed in duration-conversion task compared to other atomic tasks (25% in GPT-4 and 27% in LLaMA2_{70b}). This is because the duration-conversion task (s3)

Model	Order	Duration	Freq.	Stationarity	Typical	Avg.
GPT-4	76.4↓	92.8↑	83.3↑	71.4↓	54.5↓	77.5
GPT-3.5	50.5↑	39.8↓	55.2↑	48.4↑	28.7↓	43.5
Baichuan2 _{13b} [†]	40.5↓	51.8↑	43.7↑	46.2↑	29.8↓	42.5
LLaMA2 _{70b} [†]	65.2↑	72.1↑	66.3↑	36.3↓	52.7↓	63.0
Mistral _{7b} [†]	27.0↓	44.4↑	58.3↑	38.5↓	38.3↓	42.5

Table 6: Results in each temporal commonsense aspect under few-shot setting. Models with [†] are base models. Red ↓ and Green ↑ represent the performance is lower or higher than its average performance. Metric is EM.

necessitates a two-step reasoning process. It first unifies time units, and subsequently engages in numerical comparison. In contrast, other atomic tasks (s1, s2 and arithmetic) can be completed with a single reasoning step. In summary, LLMs perform poorly in symbolic temporal reasoning and exhibit more pronounced declines when encountering multi-step reasoning.

Mastery of commonsense knowledge varies in LLMs We analyze models’ performance across various commonsense aspects, as shown in Table 6. We regard the model’s average performance in commonsense reasoning tasks as the baseline. If the model outperforms the baseline in a specific aspect, it suggests greater proficiency in this type of knowledge, and vice versa. The findings indicate that LLMs generally demonstrate good knowledge of event duration and frequency. However, their comprehension of event order and typical events is relatively weaker. The uneven mastery of commonsense knowledge significantly affects the model’s reasoning performance, especially when dealing with complex questions that involve multiple types of knowledge. Retrieval-augmented reasoning presents a promising avenue for mitigating the model’s knowledge scarcity.

LLMs exhibit poor implicit temporal reasoning capabilities. When comparing explicit and implicit event temporal reasoning, specifically TimeQA-explicit versus others, we observe a significant performance decrease in implicit reasoning. Additionally, on TRACIE with numerous implied events, most models only surpass a random baseline (50.0). Even GPT-4, despite its advanced capabilities, achieves only a 66.4% accuracy, suggesting that the LLM struggles with modeling implicit temporal relationships. We consider it helpful to explicitly model the temporal relationships between events and time expressions, for instance constructing timelines or temporal graphs.

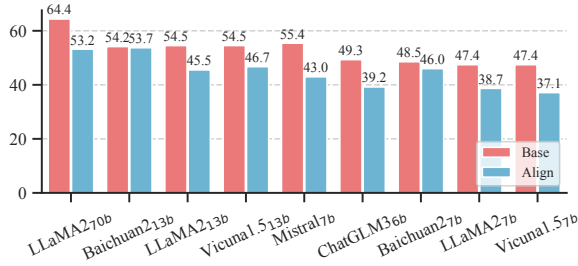


Figure 5: Performance difference between base and alignment models under few-shot setting. Baichuan2 and LLaMA2 are aligned with SFT and RLHF. Vicuna, Mistral and ChatGLM3 are aligned with only SFT.

LLMs are good factual reasoners rather than factual extractors When humans engage in temporal reasoning, it generally involves two steps: first, extracting time-fact pairs from the context, and then performing fact-based reasoning. TempReason provides extracted facts for conducting fact-based reasoning. By comparing the model’s performance in context-based (TimeQA) against fact-based (TempReason) reasoning, we identify the bottleneck in event temporal reasoning. LLMs excel in TempReason, which signifies their strong capability in fact-based reasoning. However, their performance in context-based reasoning is significantly weaker compared to their performance in fact-based reasoning. This implies that errors could arise during the extraction of time-sensitive facts from the context. We attribute this performance gap to the model’s deficiency in factual extraction capabilities. Thus, we consider LLMs to be strong factual reasoners rather than factual extractors in event temporal reasoning.

6.3 Alignment Impairs Temporal Reasoning

In the experiments mentioned earlier (Table 5), we observe a sharp decline in zero-shot performance of alignment models. To investigate whether alignment is the cause of the decline in temporal reasoning, we conducted experiments on alignment models under few-shot settings. Figure 5 illustrates the overall performance decline after alignment. With the exception of Baichuan2, all other models are severely impaired, experiencing a significant drop of up to 22%. Through manual analysis of error cases, we have summarized two reasons: (1) Alignment reduces the model’s usability, causing it to tend towards refusal to answer when confronted with knowledge-sensitive questions. (2) Alignment damages the model’s in-context learning capability,

resulting in situations where the model deviates from the demonstrations. Furthermore, we believe that the lack of temporal reasoning-related training data in alignment exacerbates this issue, leading to disparities between different reasoning capabilities, such as mathematical and temporal reasoning.

6.4 Error Analysis

We manually analyze 100 predictions by GPT-4, GPT-3.5 and LLaMA2-base_{70b} from each subtask. The visualization of errors is shown in Figure 6.

Symbolic Reasoning We categorize symbolic reasoning errors into five groups: (a) *Expression*: The model provides an incorrect time calculation expression. (b) *Computation*: The model provides the correct time calculation expression, but there is a calculation error. (c) *Conversion*: The model has an error in the conversion of time units. (d) *Comparison*: The model has an error when comparing two time-expressions (or intervals). (e) *Combination*: The model encounters errors in the combination of multiple above operations. LLMs exhibit numerous computation, conversion, and comparison errors, which suggests a substantial deficiency in their understanding of fundamental temporal expressions. Additionally, a higher frequency of errors is observed in combination questions, highlighting that multi-step reasoning continues to be a significant challenge for current models.

Commonsense Reasoning We categorize the errors of commonsense reasoning into two groups: (a) *No Answer*: The model fails to provide a final answer. (b) *Reasoning Error*: The model encounters reasoning errors, which can be subdivided into five types of knowledge-related errors. We observe that GPT series models have a higher *No Answer* rate, while LLaMA is always able to provide answers. This discrepancy can be attributed to two factors: firstly, the models may lack the necessary commonsense knowledge to formulate an answer; secondly, the preference alignment mechanism may prompt the model to abstain from answering when confronted with questions outside its knowledge scope. Integration of retrieval can alleviate the problem of knowledge scarcity to a certain degree.

Event Temporal Reasoning We categorize the errors of event temporal reasoning into four groups: (a) *No Answer*: The model is unable to find the answer in the context. (b) *Reasoning Error*: The model encounters reasoning errors. (c) *Halluci-*

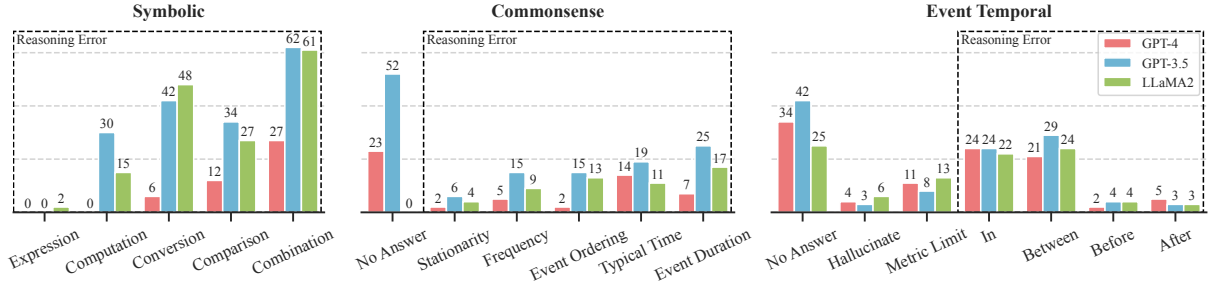


Figure 6: Error analysis for Symbolic, Commonsense, and Event Temporal. We select 100 test samples from each subtask for GPT-4, GPT-3.5 and LLaMA2-base_{70b}.

nation: The model’s prediction does not exist in the context, known as hallucination reasoning. (d) *Metric*: The model’s prediction is correct, but the metric is limited by the evaluation criteria. It can be observed that, except for reasoning errors, failures to provide answers account for approximately 30%, indicating that models still have flaws in grounding temporal facts from context. Additionally, models occasionally experience hallucination phenomena, leading to erroneous reasoning.

7 Related Work

7.1 Temporal Reasoning

There are numerous efforts addressing diverse challenges in temporal reasoning. Early research mainly relies on TimeML (Pustejovsky et al., 2003), focusing TimeX extraction and temporal relation extraction (Verhagen et al., 2007, 2010; UzZaman et al., 2013; Llorens et al., 2015; Miller et al., 2015; Mathur et al., 2021; Vashishtha et al., 2019). The advent of pre-trained language models (PLMs) has brought about commonsense reasoning as a tool to explore the world knowledge in models (Zhou et al., 2019; Qin et al., 2021; Dhingra et al., 2022). Recently, much attention has shifted towards event temporal reasoning (Chen et al., 2021; Tan et al., 2023; Wei et al., 2023). Han et al. (2021); Yang et al. (2023b); Son and Oh (2023); Chen et al. (2023) continuously pre-trains LLMs on time-aware data to elicit temporal reasoning, and Zhu et al. (2023); Su et al. (2023); Chu et al. (2023) explicitly represent temporal relationships using temporal graphs and timelines. Additionally, some works extend beyond text, evaluating temporal reasoning in structured tables and video domains (Gupta et al., 2023; Ko et al., 2023).

Some concurrent studies also analyze the temporal reasoning abilities of LLMs. Jain et al. (2023); Qiu et al. (2023) focus on temporal commonsense

and Wang and Zhao (2023) introduces a unified form for accessing the overall abilities.

Distinguished from other works, TIMEBENCH is multispectral, offering a comprehensive evaluation of LLM’s temporal reasoning abilities.

7.2 Large Language Models

In recent years, there has been rapid progress in the research of large language models (LLM) (Zhao et al., 2023). They exhibit outstanding performance across a multitude of tasks without the need for fine-tuning (Brown et al., 2020; Kojima et al., 2022). Furthermore, they have achieved astonishing results in complex reasoning tasks, such as mathematical reasoning (Cobbe et al., 2021; Mishra et al., 2022) and logical reasoning (Yu et al., 2020; Liu et al., 2023). Moreover, some studies suggest that the chain-of-thought prompting can further enhance the model’s capabilities in complex reasoning scenarios (Wei et al., 2022; Kojima et al., 2022; Chu et al., 2024; Zhang et al., 2023).

8 Conclusion

Temporal reasoning entails inherent diversity and complexity. The lack of a comprehensive benchmark makes it challenging to quantify LLMs’ temporal reasoning capabilities. In this work, we present TIMEBENCH, a comprehensive and hierarchical benchmark for LLM temporal reasoning, tailored to mirror temporal reasoning in complex scenarios. We conduct extensive experiments on state-of-the-art LLMs to investigate their temporal reasoning capabilities. Our findings indicate a substantial gap between state-of-the-art LLMs and human performance, emphasizing the need for further research in this area. Moreover, we provide a meticulous analysis and discussion, outlining the current challenges that models face and suggesting potential directions for improvement.

Limitations

TimeBench is a comprehensive benchmark to quantify the temporal reasoning capabilities of LLMs. While we have taken various factors into account, there are a few limitations. Firstly, our evaluation only applied prompt-based method under zero-shot and few-shot setting, lacking evaluations specifically tailored for models fine-tuned on the temporal domain. Secondly, the instructions and demonstrations were manually crafted, which may potentially lead to discrepancies in prompts interpretation among different LLMs. Thirdly, the dataset constituting the benchmark includes data from past years and a portion sourced from Wikipedia, which may contaminate the training corpus of LLMs.

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A TIMEBENCH Details

TIMEBENCH features 3 major categories, 10 tasks and 15 subtasks, each with distinct challenges, totaling 19,000 instances. Detailed statistics are available in Figure 7 and Table 7.

A.1 Benchmark Construction

TimeX Arithmetic (Tan et al., 2023) TimeX Arithmetic data is derived from the *11: time-time* reasoning data in TempReason. We retain 4,000 instances, where time expressions are calculated with a minimum unit of one day.

TimeX NLI (Thukral et al., 2021) The original data of TimeXNLI is in NLI format, including three sub-tasks, *Temp-Order*, *Temp-Duration*, and *Cross-Unit Duration*, including 6,140, 3,540, and 15,840 instances respectively. We conduct a random sampling of 2,213, 2,332 and 2,429 entries, resulting in a combined total of 6,965 instances.

MCTACO (Zhou et al., 2019) The original MCTACO dataset consists of yes/no questions, containing 1,332 questions with 9,442 options. To guarantee that the questions are presented in a 4-way multi-select style, we initially remove questions that have less than four options. Subsequently, to ensure that each question has at least one correct option, we filter out questions where all options are labeled as "no". For each remaining question, we randomly sample four options, striving to maintain a balance between correct and incorrect options. In most cases, a question is accompanied by 2 correct and 2 incorrect options. A minority of questions have an option distribution of 1-3 or 3-1. After the aforementioned filtering process, we obtain 852 pieces of data in a 4-way multi-select format.

DurationQA (Virgo et al., 2022) The original DurationQA has the same format as MCTACO, which consists of 694 questions with 4,868 options. Following the identical filtration procedure as MCTACO, we finally obtained a collection of 687 questions in a 4-way multi-select format.

TimeDial (Qin et al., 2021) consists of 4-way multi-select instances in a two-person dialogue scenario. We leave the original data unaltered and simply randomize the sequence of options, yielding 1,446 pieces of 4-way multi-select instances.

SituateGen (Zhang and Wan, 2023) Situated-Gen includes 1,220 test cases, which span across two distinct reasoning domains: *time* and *geography*. We manually screen the original test data and retain those with clear time features for temporal reasoning evaluation, resulting in 115 instances.

TimeQA (Chen et al., 2021) The original data of TimeQA includes two splits, *Easy* and *Hard*, with each question containing 20 Wikipedia paragraphs. The excessively long context may exceed the model’s maximum length limit and incur significant inference overhead. Therefore, we have reduced the context of the original data. For the paragraphs in the original data, we refer to those containing the answer as relevant paragraphs, and the rest as irrelevant paragraphs. For each question, we keep the first paragraph, all relevant paragraphs, and one random irrelevant paragraph as distractor. This ensures that most questions have at least three paragraphs. After that, we sample 500 pieces of data from those where the context length is less than 650 tokens. For both *Easy* and *Hard* splits,

we apply the aforementioned filtration, resulting in 500 questions each, totaling 1,000 instances.

TempReason (Tan et al., 2023) TempReason dataset contains 5,397 entries for 12 (event-time reasoning) and 4,426 entries for 13 (event-event reasoning). In the original dataset, each question corresponds with a text context and extracted facts. Similar to TimeQA, we apply a filter based on context length. We preserve questions with a context length between 300 and 600 tokens, yielding 839 and 1,037 instances, respectively. Notably, every remaining question is applicable to either context-based reasoning or fact-based reasoning.

MenatQA (Wei et al., 2023) MenatQA consists of 999 data entries, formatted similarly to TimeQA, where each question is accompanied by several corresponding paragraphs. Following the paper’s proposed method, we modify the original data by incorporating the three time-sensitive factors: scope, order, and counterfactual. Subsequently, for each factor, we randomly sample 400 instances, resulting in a total of 1,200 data points.

TRACIE (Zhou et al., 2021) The original TRACIE dataset consists of yes/no type questions, containing 4,248 test instances. We randomly sample 500 instances from the *iid* split in the test set.

A.2 Human Performance Evaluation

Unless otherwise stated, the results of human evaluation are derived from original dataset papers. Please refer to the corresponding paper for human evaluation details. TimeXNLI, Date Arith, and MCTACO are manually evaluated by three authors from the TimeBench team. Within each subtask, we randomly sample 50 instances, and the average of the performances by three human evaluators is considered the final human performance.

A.3 Task Formats

TIMEBENCH is a multispectral benchmark, which features four different task formats.

Multi-Select Questions Previous work utilizes the Multiple Choice (MC) form, which requires models to select the only correct answer from the options. However, this task form has shortcuts and may not truly reflect the model’s abilities. To address this, we employ the Multi-Select (M-S) task form, where the model needs to select all possible correct answers from the options provided. In our

task, each question presents four options, with at most two of them being correct.

Natural Language Inference is the task of determining the logical relationship between two pieces of text. Specifically, given a premise and a hypothesis, the model needs to determine whether the hypothesis can be inferred from the premise and output entailment, contradiction, or neutral. Our tasks focus on the entailment in temporal domains.

Free-form Reading Comprehension requires models to answer questions based on the provided context, and the ground truth answer is free-form without pre-defined format restrictions.

Constrained Text Generation refers to the task of generating text under certain constraints. The task is keyword-constrained text generation, where the model takes keywords as input and outputs sentences that include those keywords.

A.4 Evaluation Metrics

Accuracy is used for NLI and date arithmetic tasks. M-S tasks are evaluated using option-level EM and F1. FRC tasks (excluding date arithmetic) are assessed with token-level EM and F1. For CTG task, we take the average of multiple generation metrics, which are outlined as follows.

Metrics for SituatedGen Following SituatedGen (Zhang and Wan, 2023), we use BLEU-4 (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), ROUGE-L (Lin, 2004), CIDEr (Vedantam et al., 2015), and MATCH (Zhang and Wan, 2023) scores to metric the results of CTG.³

The overall score is calculated as the sum of the above scores. We set the weight of CIDEr to 1/10 for balancing when summation.

$$S = \text{BLEU-4} + \text{METEOR} + \text{ROUGE-L} + \text{CIDEr}/10 + \text{MATCH}$$

As the overall score S does not represent a percentile, we proceed to normalize the models’ scores to align with humans’ relative performance levels.

B Supplemental Materials

B.1 Models

ChatGPT-3.5/GPT-4 (Ouyang et al., 2022; OpenAI, 2023) ChatGPT is a chat model aligned

³We utilize [pycocoevalcap](#) package to calculate BLEU-4, METEOR, ROUGE-L, CIDEr.

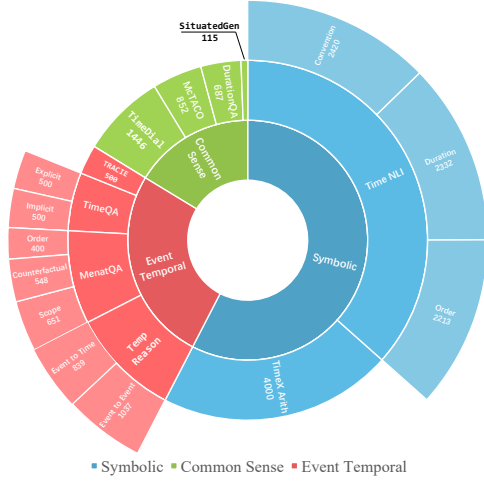


Figure 7: The quantity and proportion of data for each task and its respective subtasks within TIMEBENCH.

through SFT and RLHF based on GPT-3 (Brown et al., 2020). GPT-4 is an upgraded version of ChatGPT with enhanced reasoning capabilities, making it the most powerful LLM. Unless otherwise stated, ChatGPT refers to *gpt-3.5-turbo-0613* and GPT-4 refers to *gpt-4-0613*.

Llama2/Vicuna-1.5 (Touvron et al., 2023; Chiang et al., 2023) LLaMA2 is an open foundation model trained on 2T tokens with efficient grouped-query attention (Ainslie et al., 2023). LLaMA2-chat is the official aligned model with SFT and RLHF, and Vicuna-1.5 is aligned with SFT only by the community⁴.

Baichuan2 (Yang et al., 2023a) is an open foundation model pre-trained on 2.6T tokens, which is competitive with LLaMA2. Baichuan2-chat is the official aligned model with SFT and RLHF.

Mistral (Jiang et al., 2023) is a 7B open foundation model incorporating efficient grouped-query attention (Ainslie et al., 2023) and sliding windows attention (Beltagy et al., 2020). It achieves the strongest performance among models of its size, even surpassing LLaMA2-13B. Mistral-instruct is the officially aligned model with SFT only.

ChatGLM3 (Zeng et al., 2023) is an open-source bilingual LLM for Chinese and English, exhibiting competitive performance under 10B.

FLAN-T5 (Chung et al., 2022) is an open-source instruction model built on top of T5 (Raffel et al., 2020) through instruction fine-tuning.

⁴<https://lmsys.org/>

B.2 Full Results

The overall score is derived from the average of all corresponding metrics. For brevity, we omit some F1 scores in tables in the main text. Please refer to Table 9 for the full experimental results. The full results of SituatedGen can be found in Table 8.

B.3 Prompts

The prompt formats are showcased in Figure 9. The demonstrations can be found from Figure 10 to 18.

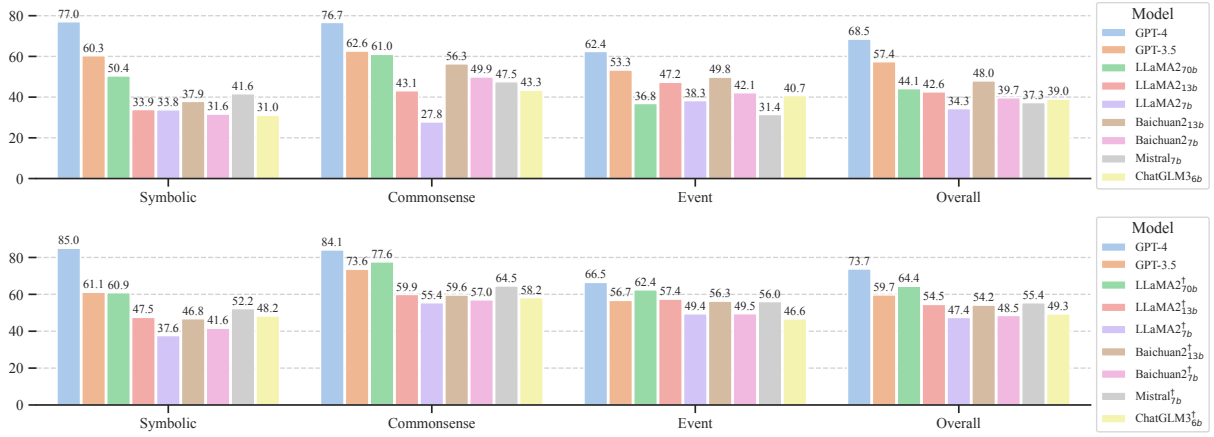


Figure 8: Performance comparison between state-of-the-art LLMs. **Up:** GPT-4/3.5 and alignment models under zero-shot setting. **Down:** GPT-4/3.5 and base models under few-shot setting.

Dataset	Format	#	Challenges
Symbolic			
TimeX Arith	FRC	4,000	TimeX Arithmetic
TimeX NLI	NLI	6,965	TimeX Causality
- <i>Order</i>	-	2,213	order
- <i>Duration</i>	-	2,332	duration
- <i>Conversion</i>	-	2,420	duration + time unit conversion
Commonsense			
MCTACO	M-S	852	Temporal Commonsense
TimeDial	M-S	1,446	Temporal Commonsense
DurationQA	M-S	687	Event Duration
SituatedGen	CTG	115	Temporal Commonsense
Event			
TimeQA	FRC	1,000	Context-based Reasoning
- <i>Explicit</i>	-	500	explicit, event-time reasoning
- <i>Implicit</i>	-	500	implicit, event-time reasoning
MenatQA	FRC	1,599	Implicit, Context-based Reasoning
- <i>Order</i>	-	400	event-time reasoning
- <i>Scope</i>	-	400	event-time reasoning
- <i>Counterfactual</i>	-	400	event-time reasoning
TempReason	FRC	1,876	Implicit, Fact-based Reasoning
- <i>l2 (e2t)</i>	-	839	event-time reasoning
- <i>l3 (e2e)</i>	-	1,037	event-event reasoning
TRACIE	NLI	500	Implicit, Implied Event-Event Reasoning
<i>In total</i>		19,000	

Table 7: The statistics, task formats and challenges in TIMEBENCH.

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	MATCH	Overall	Norm
Human	39.9	40.4	56.3	397	98.1	274.4	100.0
GPT-4 + FS	8.23 28.64	31.27 38.99	28.84 55.69	38.45 298.64	90.41 90.11	162.59 243.29	59.25 88.66
GPT-3.5 + FS	13.38 27.24	30.12 33.77	35.91 51.18	125.41 282.75	78.76 76.54	170.70 217.01	62.21 79.08
LLaMA2 _{70b} + FS	5.15 19.10	13.62 29.09	15.83 41.74	22.07 171.36	31.79 65.29	68.60 172.35	25.00 62.81
LLaMA2 _{13b} + FS	4.66 15.15	21.43 27.49	20.80 37.55	17.72 138.13	61.62 64.94	110.28 158.93	40.19 57.92
LLaMA2 _{7b} + FS	2.77 6.90	13.46 15.82	14.69 21.77	14.34 52.99	34.83 33.81	67.18 83.60	24.48 30.47
Baichuan2 _{13b} + FS	8.33 15.79	25.86 30.23	30.07 40.96	82.63 169.14	70.63 71.01	143.15 174.91	52.17 63.74
Baichuan2 _{7b} + FS	5.17 15.06	21.99 23.45	23.73 32.29	44.80 137.94	59.85 52.04	115.22 136.64	41.99 49.79
Vicuna1.5 _{13b} + FS	7.73 6.85	26.35 18.66	29.15 25.99	69.16 92.96	71.91 46.19	142.06 106.99	51.77 38.99
Vicuna1.5 _{7b} + FS	6.29 20.71	24.34 30.19	26.91 45.20	46.90 203.20	68.84 67.58	131.07 184.00	47.77 67.05
FLAN-T5 + FS	16.20 12.88	24.43 30.38	29.38 36.27	95.17 92.20	56.38 76.44	135.91 165.19	49.53 60.20
Mistral _{7b} + FS	5.82 18.96	22.89 29.02	24.19 43.15	44.03 185.61	63.74 63.24	121.03 172.93	44.11 63.02
ChatGLM3 _{6b} + FS	6.56 10.53	21.11 24.17	21.96 33.44	41.48 124.50	53.02 56.94	106.80 137.53	38.92 50.12
LLaMA2 _{70b} [†]	22.34	33.03	50.93	243.31	74.96	205.59	74.92
LLaMA2 _{13b} [†]	17.54	29.44	45.21	200.14	65.64	177.84	64.81
LLaMA2 _{7b} [†]	17.49	28.33	45.24	202.08	59.98	171.25	62.41
Baichuan2 _{13b} [†]	17.86	29.75	44.28	198.83	66.35	178.12	64.91
Baichuan2 _{7b} [†]	15.30	27.54	41.80	171.59	62.40	164.20	59.84
Mistral _{7b} [†]	14.54	27.39	41.72	168.89	59.42	159.96	58.30
ChatGLM3 _{6b} [†]	17.11	29.35	40.74	156.49	66.18	169.02	61.60

Table 8: Full results of SituatedGen. Aligned models are under zero-shot setting by default. The top-3 results are **bold**. Methods with [†] are base models without alignment, under few-shot setting. We consider human performance as 100 points and normalize models’ results accordingly.

Method	Symbolic			Commonsense										Event										Overall						
	TimeXNLI		Date Arith	DurationQA		McTACO		TimeDial		SitGen	TimeQA		MenatQA		TempReason		TRACIE	Sym.	Comm.	Event Avg.										
	s1	s2		EM	F1	EM	F1	EM	F1		E-EM	E-F1	H-EM	H-F1	S-EM	S-F1					O-EM	O-F1	C-EM	C-F1	L2-EM	L2-F1	L3-EM	L3-F1	Acc	
Human	98.0	96.0	92.0	100.0	64.0	80.8	75.8	87.1	97.8	97.8	100.0	89.0	93.3	87.0	91.1	82.0	85.6	84.0	87.3	76.0	79.9	96.0	97.1	94.0	95.3	82.5	96.5	91.4	89.0	91.5
GPT-4	78.6	76.0	50.7	98.0	35.0	59.2	61.2	80.0	72.0	91.1	59.3	48.9	60.6	40.4	46.5	44.4	57.0	49.0	57.0	22.0	23.1	91.0	95.3	94.0	95.0	64.8	75.8	72.4	62.4	68.3
+ CoT	80.0	76.0	60.0	92.0	35.0	58.1	67.0	82.6	65.0	89.3	-	50.0	61.3	33.0	41.2	43.4	54.6	53.0	59.6	20.0	22.6	93.0	97.0	93.0	94.5	58.0	77.0	76.7	61.1	68.5
+ FS	85.3	73.3	53.3	100.0	51.0	64.8	77.6	88.3	85.0	94.6	88.6	59.2	73.7	40.0	51.0	59.6	72.4	48.0	54.8	25.3	28.7	86.0	92.4	94.8	95.9	62.8	78.0	84.1	66.5	73.7
+ FS CoT	92.0	84.0	64.0	100.0	42.0	55.1	68.0	72.3	79.0	93.4	-	48.0	66.9	44.4	52.8	48.5	65.3	44.0	52.6	22.0	25.9	91.0	96.9	93.0	94.6	66.4	85.0	73.6	65.2	72.1
GPT-3.5	45.4	67.6	31.2	97.0	19.2	50.5	34.1	68.6	39.2	69.1	62.3	60.5	70.8	29.5	35.4	36.5	40.9	37.5	43.9	21.0	22.9	73.6	81.2	61.8	73.8	57.4	60.3	62.6	53.3	57.4
+ CoT	33.6	64.8	33.6	71.0	12.4	23.2	28.1	45.1	34.6	67.0	-	52.5	64.4	29.0	35.1	35.8	39.7	38.5	42.9	24.0	26.3	32.0	57.6	54.2	68.1	52.0	50.8	45.1	48.3	48.3
+ FS	52.0	68.4	31.6	63.6	42.8	67.7	43.5	71.2	47.8	76.4	79.1	53.8	66.1	37.9	48.4	37.8	43.2	43.5	51.6	16.0	17.9	77.7	84.7	70.0	78.0	55.0	53.9	73.6	55.6	59.7
+ FS CoT	51.6	71.8	36.6	84.4	20.8	41.2	21.4	38.1	48.3	71.1	-	56.5	68.0	37.5	47.0	38.1	42.5	37.5	41.7	33.0	37.8	86.2	89.9	68.0	76.6	50.2	61.1	50.1	56.7	56.6
LLaMA2-70b	44.0	47.0	32.0	78.5	12.7	59.2	23.0	68.9	10.0	57.0	25.0	28.0	40.8	31.0	40.6	8.0	18.9	11.0	16.6	9.0	12.0	50.0	63.5	39.0	54.5	50.4	50.4	52.5	36.8	44.1
+ CoT	30.0	66.0	28.0	53.5	8.0	57.3	21.0	67.1	9.0	58.6	62.8	17.0	31.4	13.0	19.5	5.0	12.2	8.0	12.7	18.0	20.8	12.0	37.5	20.0	40.5	51.0	44.4	61.0	28.2	39.1
+ FS	49.0	42.0	38.0	62.0	1.3	61.2	13.0	66.5	6.0	56.6	62.8	41.0	51.1	16.0	20.0	8.0	16.4	17.0	19.9	18.0	18.7	34.0	52.2	31.0	41.1	51.0	47.8	61.8	33.8	44.3
+ FS CoT	54.0	63.0	40.0	69.5	8.0	55.2	21.5	62.1	6.0	56.4	-	36.6	50.9	34.0	42.4	28.0	38.6	19.0	29.3	18.0	21.9	77.0	83.1	65.0	74.7	57.0	56.6	57.9	49.7	53.2
LLaMA2-136b	30.0	49.0	34.0	22.5	4.0	38.5	8.5	40.6	10.0	35.4	57.9	46.0	61.9	21.0	30.5	28.0	46.1	23.0	36.1	18.0	26.9	43.0	53.1	55.0	69.4	49.0	33.9	43.1	46.6	42.6
+ CoT	36.0	50.0	38.0	6.0	7.3	39.2	14.0	51.7	10.0	36.9	-	45.0	58.7	30.0	38.9	20.0	40.9	18.0	32.5	21.0	33.6	43.0	58.0	56.0	68.4	47.0	32.5	42.6	47.3	42.4
+ FS	43.0	57.0	60.0	20.5	9.0	46.8	8.0	66.6	15.0	62.3	40.2	24.0	34.2	17.0	18.4	11.0	25.9	5.0	14.6	22.0	33.3	54.0	68.1	50.0	64.8	47.0	45.1	54.0	38.3	43.9
+ FS CoT	37.0	55.0	50.0	33.0	12.0	49.5	11.0	45.6	8.0	44.5	-	35.0	46.0	21.0	25.4	34.0	46.7	23.0	36.5	7.0	16.5	72.0	80.8	54.0	66.2	50.0	43.8	46.5	46.0	45.5
LLaMA2-7b	39.0	53.0	30.0	13.0	2.7	39.3	4.0	41.0	1.0	6.3	24.5	37.0	49.0	14.0	29.0	7.0	26.8	8.0	21.1	9.0	16.0	48.0	63.9	32.0	47.9	49.0	33.8	27.8	37.8	34.3
+ CoT	44.0	50.0	33.0	5.0	2.7	35.0	4.5	40.0	1.0	1.7	-	27.0	49.0	17.0	31.6	11.0	31.4	10.0	24.5	7.0	17.8	44.0	56.9	32.0	48.1	46.0	33.0	25.6	38.3	34.3
+ FS	44.0	60.0	34.0	11.0	4.0	62.8	8.0	64.7	8.0	40.0	30.5	36.0	50.8	20.0	29.4	5.0	22.3	6.0	18.0	6.0	17.9	12.0	36.3	23.0	44.3	53.0	37.3	49.5	34.0	38.7
+ FS CoT	38.0	51.0	36.0	14.5	11.0	42.8	25.0	65.6	13.0	53.4	-	36.0	53.5	21.0	34.1	1.0	13.6	3.0	11.2	5.0	14.0	22.0	46.7	21.0	42.3	51.0	34.9	53.9	33.3	37.8
Baichuan2-136b	41.0	61.0	37.0	12.5	4.0	52.0	18.5	63.4	15.0	57.7	52.2	45.0	55.4	29.0	34.6	31.0	48.8	34.0	44.3	30.0	39.5	40.0	57.4	45.0	61.4	49.0	37.9	56.3	48.8	48.0
+ CoT	40.0	57.0	31.0	10.0	3.3	44.6	20.0	61.9	13.0	38.1	-	36.0	41.5	36.0	40.9	39.0	52.0	27.0	38.5	29.0	43.2	46.0	64.3	55.0	48.0	55.0	34.5	54.9	49.8	46.7
+ FS	43.0	59.0	40.0	42.5	24.7	62.1	27.5	70.2	18.0	58.9	63.7	47.0	60.7	30.0	42.1	27.0	37.8	23.0	35.7	10.0	20.4	40.0	57.4	37.0	53.0	51.0	36.5	57.3	44.8	45.8
+ FS CoT	45.0	54.0	48.0	47.0	10.7	44.4	27.0	68.8	15.0	55.0	-	43.0	57.8	27.0	36.7	38.0	49.8	34.0	40.7	33.0	43.0	72.8	80.4	43.0	60.2	44.0	48.5	56.1	51.6	51.7
Baichuan2-7b	35.0	50.0	37.0	4.5	4.0	47.9	10.5	55.3	15.0	54.3	42.0	26.0	41.5	20.0	34.7	20.0	35.2	19.0	31.2	6.0	20.4	22.0	43.4	29.0	47.7	55.0	31.6	49.9	38.6	39.7
+ CoT	38.0	43.0	32.0	1.0	5.3	37.9	13.0	58.0	15.0	44.2	-	41.0	53.5	28.0	38.8	29.0	39.9	23.0	33.2	18.0	29.3	21.0	41.2	29.0	47.2	54.0	28.5	46.7	42.1	39.4
+ FS	40.0	50.0	36.0	20.0	28.7	59.4	26.5	66.9	17.0	53.0	49.8	45.0	60.7	30.0	42.1	27.0	37.8	23.0	35.7	10.0	20.4	40.0	57.4	37.0	53.0	51.0	36.5	57.3	44.8	45.8
+ FS CoT	41.0	50.0	36.0	23.5	13.0	45.7	17.5	58.1	7.0	39.2	-	36.0	51.2	29.0	43.0	42.0	52.5	25.0	39.3	20.0	31.0	57.0	70.1	39.0	60.2	49.0	37.6	47.7	49.5	46.0
Vicuna1.5-136b	35.0	50.0	36.0	15.0	8.0	39.2	21.5	59.1	7.0	34.2	51.8	43.0	60.4	29.0	37.0	38.0	46.8	22.0	37.4	17.0	23.2	14.0	42.1	13.0	43.6	46.0	34.0	46.1	42.1	41.1
+ CoT	42.0	51.0	37.0	3.0	1.3	29.8	11.5	50.0	7.0	33.7	-	44.0	56.9	31.0	36.4	16.0	38.2	25.0	37.7	13.0	20.4	31.0	49.0	29.0	49.1	51.0	33.3	37.8	42.3	39.0
+ FS	48.0	57.0	38.0	30.5	7.3	33.6	27.5	57.0	13.0	40.3	39.0	45.0	58.3	23.0	25.9	38.0	42.6	26.0	41.4	18.0	20.1	51.0	61.8	28.0	42.6	56.0	43.4	42.5	43.6	43.3
+ FS CoT	38.0	59.0	39.0	39.5	10.7	37.4	14.0	45.8	12.0	41.6	-	47.0	59.5	27.0	30.7	39.0	48.1	31.0	35.9	26.0	31.2	71.0	77.5	53.0	65.5	52.0	43.9	41.6	50.1	46.7
Vicuna1.5-7b	37.0	58.0	43.0	5.0	1.3	40.4	9.5	52.5	6.0	32.0	47.8	35.0	47.1	11.0	18.5	20.0	35.7	15.0	25.7	12.0	17.3	14.0	33.0	14.0	46.8	54.0	35.8	43.2	34.8	37.1
+ CoT	36.0	50.0	36.0	1.5	1.3	39.4	8.5	49.2	9.0	36.2	-	30.0	40.9	14.0	24.6	16.0	26.2	14.0	28.5	12.0	25.0	9.0	27.7	7.0	40.3	54.0	30.9	41.6	33.4	34.4
+ FS	43.0	57.0	37.0	8.5	3.3	44.6	5.5	42.1	7.0	36.8	67.1	24.0	31.9	12.0	14.9	16.0	21.8	20.0	27.5	17.0	22.2	13.0	34.3	6.0	32.2	54.0	36.4	47.7	29.9	35.9
+ FS CoT	35.0	54.0	35.0	8.0	2.7	3																								

Method	Symbolic			Commonsense										Event										Overall														
	TimeXNLI			Date Arith		DurationQA		McTACO		TimeDial		SitGen		TimeQA				MenatQA				TempReason				TRACIE		Sym.	Comm.	Event	Avg.							
	s1	s2	s3	Acc	EM	F1	EM	F1	EM	F1	EM	F1	Norm	E-EM	E-F1	H-EM	H-F1	S-EM	S-F1	O-EM	O-F1	C-EM	C-F1	L2-EM	L2-F1	L3-EM	L3-F1					Acc						
+ FS CoT	29.0	62.0	32.0	28.5			9.3	46.6	14.5	49.2	8.0	34.4	-			13.0	33.7	11.0	25.3	9.0	26.4	9.0	24.3	9.0	20.3	68.0	78.6	42.0	57.4	50.0	37.9	43.4	39.5	39.8				
	ChatGLM3 _{6b}																																					
	38.0	50.0	34.0	2.0			3.0	34.1	7.0	43.6	14.0	56.7	38.9			20.0	41.2	14.0	31.7	25.0	33.8	17.0	26.0	24.0	32.2	42.0	57.0	30.0	54.0	50.0	31.0	43.3	40.7	39.0				
	27.0	49.0	37.0	0.0			1.0	24.8	3.0	37.1	10.0	44.8	-			24.0	41.7	10.0	25.4	27.0	34.6	22.0	28.1	35.0	41.2	28.0	44.5	27.0	52.0	48.0	28.3	35.6	39.4	35.7				
+ FS	37.0	52.0	30.0	0.0			10.0	53.0	9.0	52.9	19.0	54.7	50.1			11.0	19.6	13.0	17.0	25.0	35.0	24.0	30.4	20.0	25.6	27.0	46.5	33.0	53.3	54.0	29.8	52.7	35.2	38.2				
	32.0	66.0	31.0	0.0			4.0	34.8	8.0	43.6	11.0	44.0	-			25.0	43.3	19.0	27.5	23.0	30.2	19.0	25.7	36.0	43.0	50.0	62.8	34.0	56.0	48.0	32.3	40.8	42.1	39.2				
LLaMA2-Base _{70b} + CoT	55.0	61.0	37.0	82.0			40.0	67.4	59.0	85.3	63.0	82.7	74.9			57.0	66.7	36.0	48.3	52.0	61.4	35.0	42.5	25.0	33.8	78.0	83.2	80.0	85.4	61.0	58.8	77.6	60.5	64.4				
	52.0	73.0	39.0	79.5			32.6	62.3	47.0	79.1	25.0	61.1				55.0	64.3	34.0	43.0	50.0	57.7	37.0	45.2	44.0	53.1	82.0	87.5	76.0	81.6	67.0	60.9	67.5	62.4	63.0				
LLaMA2-Base _{13b} + CoT	50.0	54.0	30.0	29.5			19.3	53.3	26.5	66.0	20.0	55.6	64.8			48.0	59.3	34.0	48.6	41.0	49.6	38.0	43.4	34.0	37.5	68.0	78.7	49.0	62.7	58.0	40.9	59.9	54.7	52.6				
	40.0	61.0	37.0	52.0			25.3	59.3	26.0	68.8	11.0	40.8	-			46.0	59.4	37.0	49.1	49.0	58.4	34.0	43.8	38.0	44.1	70.0	78.0	55.0	68.2	58.0	47.5	56.3	57.4	54.5				
LLaMA2-Base _{7b} + CoT	26.0	50.0	30.0	20.0			19.3	54.5	20.0	59.6	15.0	45.2	62.4			44.0	54.4	30.0	45.3	42.0	49.8	34.0	41.9	30.0	35.8	50.0	64.0	36.0	53.3	49.0	31.5	55.4	49.2	46.3				
	37.0	52.0	36.0	25.5			21.3	56.9	26.5	67.0	16.0	41.9	-			32.0	45.6	27.0	36.1	41.0	50.9	30.0	38.0	51.0	57.3	45.0	59.7	37.0	57.7	50.0	37.6	55.3	49.4	47.4				
Baichuan2-Base _{13b} + CoT	38.0	48.0	33.0	42.5			20.7	54.8	42.5	73.0	11.0	45.7	64.9			50.0	59.4	40.0	54.2	42.0	52.7	31.0	38.0	13.0	21.4	68.0	77.3	50.0	63.5	54.0	40.4	59.6	52.6	51.3				
	50.0	56.0	34.0	47.0			29.3	62.0	22.5	69.3	12.0	43.8	-			46.0	58.2	39.0	49.6	39.0	49.8	34.0	40.1	37.0	45.6	73.0	81.3	46.0	65.6	60.0	46.8	58.4	56.3	54.2				
Baichuan2-Base _{7b} + CoT	27.0	66.0	41.0	32.5			28.0	59.8	34.5	69.4	5.0	34.3	59.8			40.0	53.8	35.0	50.2	41.0	49.6	33.0	38.5	18.0	22.9	49.0	65.9	34.0	51.0	55.0	41.6	55.8	48.4	48.5				
	30.0	56.0	34.0	34.0			23.3	57.0	33.0	69.5	12.0	44.5	-			41.0	51.2	31.0	40.7	38.0	46.4	26.0	32.6	41.7	46.3	46.0	61.5	43.8	64.1	53.0	38.5	57.0	49.5	48.1				
Mistral-Base _{7b} + CoT	48.0	53.0	38.0	41.0			34.0	61.8	42.5	76.2	35.0	61.8	58.3			43.0	55.9	30.0	45.3	37.0	49.4	38.0	47.8	37.0	45.5	68.0	76.7	64.0	74.8	53.0	45.0	64.5	56.1	55.4				
	57.0	63.0	35.0	54.0			30.0	61.8	42.0	45.7	29.0	57.3	-			51.0	60.4	30.0	46.2	48.0	57.2	37.0	47.9	24.0	33.2	60.0	65.9	58.0	67.9	57.0	52.3	54.9	54.5	54.0				
ChatGLM3-Base _{6b} + CoT	48.0	70.0	32.0	35.0			3.3	51.8	13.5	62.6	11.0	55.0	61.6			50.0	57.2	24.0	26.3	30.0	35.4	38.0	41.5	22.0	22.5	67.0	76.4	35.0	55.9	58.0	46.3	57.8	46.7	49.3				
	47.0	68.0	32.0	46.0			8.7	53.9	15.5	64.3	13.0	56.5				45.0	52.5	23.0	24.5	30.0	35.0	37.0	40.2	22.0	22.5	72.0	79.4	42.0	60.3	54.0	48.3	58.2	46.1	49.1				

Table 9: Full results of TimeBench. Aligned models are under zero-shot setting by default. Methods with † are base models without alignment, under few-shot setting, thus incomparable with other methods. We consider human performance as 100 points and normalize models’ results accordingly.

DURATIONQA, MCTACO
Answer the following question, select all the possible correct options, and each question has at least one correct option. Context: {} Question: {} Options: {} Answer:
TIME DIAL
There is a two-person dialogue with several options. Choose all appropriate options to substitute the <mask> in the dialogue, and each question has at least one correct option. Dialogue: {} Options: {} Answer:
TRACIE
Read the following story and hypothesis, determine whether the hypothesis can be inferred from the story. You need to understand the implicit temporal relationships between events to make judgments. Story: {} Hypothesis: {} Options: A. Entailment B. Contradiction Answer:
SITUATEDGEN
Generate a pair of contrastive sentences with the given set of keywords. Keywords: {}
DATE ARITHMETIC
Question: {}? Answer:
TIMEQA
I will give you a question with context. You need to answer my question based on the context. If you can infer the answer from the context, then output your answer. Otherwise, if there is no answer, output [unanswerable]. Context: {} Question: {} Answer:
TEMPREASON
I will give you a question with context. You need to answer my question based on the context. Context: {} Question: {} Answer:
MENATQA
Get answers for the question based on the context, where answers derived from substrings in the context or categorized as [unanswerable]. Context: {} Question: {} Answer:
TIME X-NLI
Read the following statements about time and determine if the hypothesis can be inferred from the premise. Premise: {} Hypothesis: {} Options: A. Entailment B. Contradiction C. Neutral Answer:

Figure 9: Zeroshot instructions and input formats.

CoT Demonstration of TIMEX-NLI (3-shot, order)
Answer the following question, select all the possible correct options, and each question has at least one correct option. Premise: On Wednesday, they got married. Hypothesis: Before Friday, they got married. Options: A. Entailment B. Contradiction C. Neutral Answer: Wednesday is before Friday. As a result, we can infer that if something happens on Wednesday, it definitely happens before Friday. Therefore, the answer is A. Entailment. Premise: We went to Disneyland on Monday. Hypothesis: We went to Disneyland after Wednesday. Options: A. Entailment B. Contradiction C. Neutral Answer: Monday is before Wednesday. As a result, We can infer that if something happens on Monday, it definitely can not happen after Wednesday. Therefore, the answer is B. Contradiction. Premise: The failing company issued major layoffs after Tuesday. Hypothesis: The failing company issued major layoffs after Thursday. Options: A. Entailment B. Contradiction C. Neutral Answer: Tuesday is before Thursday. If something happened after Tuesday, we cannot be certain whether it occurred after Thursday. Therefore, the answer is C. Neutral.

Figure 10: Chain-of-Thought demonstrations of TimeX-NLI (s1-order).

CoT Demonstration of DATE ARITHMETIC (4-shot)
Question: What is the time 4 year and 1 month after Apr, 2000? Answer: First, 4 years after 2000 is 2004. Next, 1 month after April is May. Therefore, 4 year and 1 month after Apr, 2000 is May, 2004. Question: What is the time 3 year and 4 month before Jun, 1840? Answer: First, subtracting 3 years from 1840 gives 1837. Next, subtracting 4 months from June gives February. Therefore, 3 year and 4 month before Jun, 1840 is Feb, 1837. Question: What is the time 7 year and 11 month after Feb, 1819? Answer: First, 7 years after 1819 is 1826. Next, 11 months after February is January of the next year. Therefore, 7 years and 11 months after Feb, 1819 is Jan, 1827. Question: What is the time 6 year and 9 month before Jan, 1234? Answer: First, subtracting 6 years from 1234 gives 1228. Next, subtracting 9 months from January gives April of the previous year. Therefore, 6 year and 9 month before Jan, 1234 is Apr, 1227.

Figure 11: Chain-of-Thought demonstrations of Date Arithmetic.

CoT Demonstration of TRACIE (4-shot)
Read the following story and hypothesis, determine whether the hypothesis can be inferred from the story. You need to understand the implicit temporal relationships between events to make judgments Story: Joe was a police officer. Joe was patrolling the streets of the city in his cruiser. Suddenly, Joe was alerted to a crime happening near him by dispatch. Joe responded to the scene and found a bank robber fleeing on foot. Joe arrested the criminal and was promoted. Hypothesis: Joe put on his police uniform. starts after Joe arrest the criminal Options: A. Entailment B. Contradiction Answer: From the story we know Joe was patrolling. In the work state, Joe has already put on the police uniform. So we can infer that Joe put on his police uniform before arresting the criminal. This conflicts with hypothesis. Therefore, the answer is B. Contradiction.

Figure 12: Chain-of-Thought demonstrations of TRACIE.

CoT Demonstration of DURATIONQA (4-shot)
Answer the following question, select all the possible correct options, and each question has at least one correct option. Context: actually i have an project on it so please give me as much as you have information about migratory birds in punjab Question: How long did it take for them to have information about migratory birds in punjab? Options: A. several months B. 12 weeks C. a few minutes D. almost instantly Answer: This is a conversation scenario. In the conversation, providing relevant information about migratory birds in punjab to him is in real-time and takes very little time. Therefore, the answer is C. a few minutes, D. almost instantly. Context: Hope she stops laying eggs because she will get really skinny ! Question: How long did it take for her to lay eggs? Options: A. 1 week B. 22 hours C. 2 years D. 4 years Answer: According to commonsense knowledge, the time it takes for birds to lay eggs typically varies from one day to several days. Therefore, the answer is A. 1 week, B. 22 hours.

Figure 13: Chain-of-Thought demonstrations of DurationQA.

CoT Demonstration of MCTACO (4-shot)
Answer the following question, select all the possible correct options, and each question has at least one correct option. Context: She ordered the tastiest kind of each vegetable and the prettiest kind of each flower. Question: How often does she order vegetables and flowers? Options: A. once a second B. three days a week C. every 10 centuries D. once a week Answer: According to commonsense knowledge, ordering vegetables and flowers typically happens on a regular basis, usually every few days. Therefore, the answer is B. three days a week, D. once a week. Context: Wallace, 38, called Gastonia home from the age of 8 until she graduated from Hunter Huss High School in 1983. Question: When did Wallace wake up for high school? Options: A. at 6 am B. at 1 am C. 7:00 AM D. at 6 pm Answer: According to commonsense knowledge, waking up for high school typically happens in the morning, usually between 6 AM and 8 AM. Therefore, the answer is A. at 6 am, C. 7:00 AM.

Figure 14: Chain-of-Thought demonstrations of MCTACO.

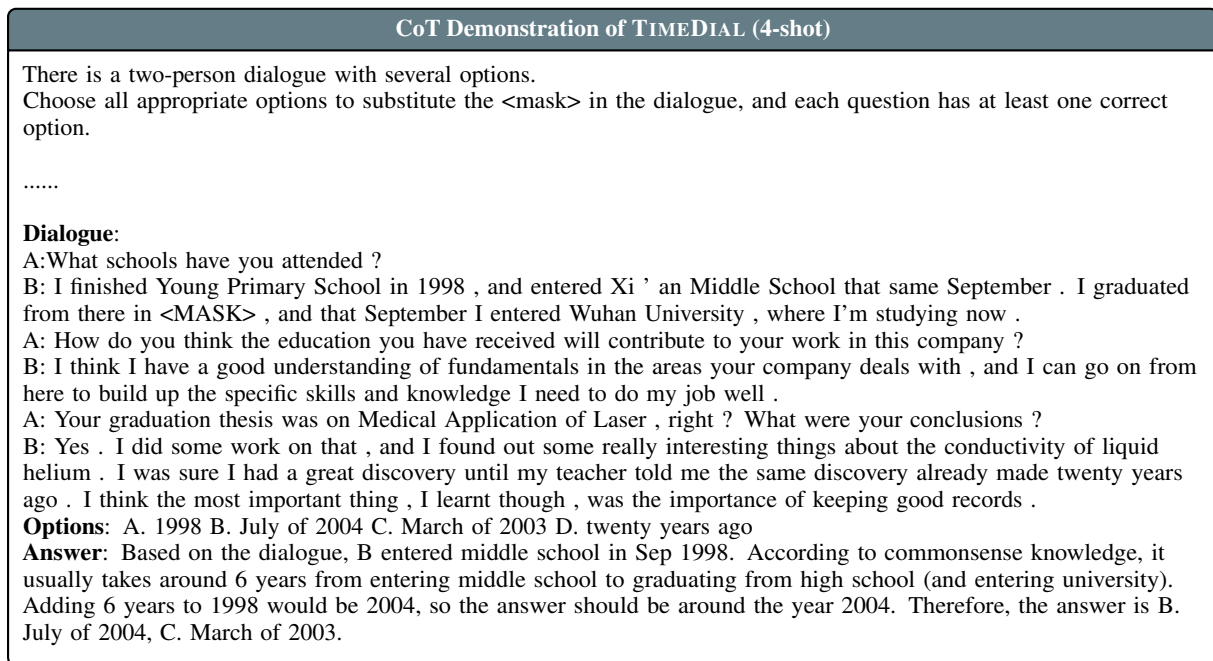


Figure 15: Chain-of-Thought demonstrations of TimeDial.

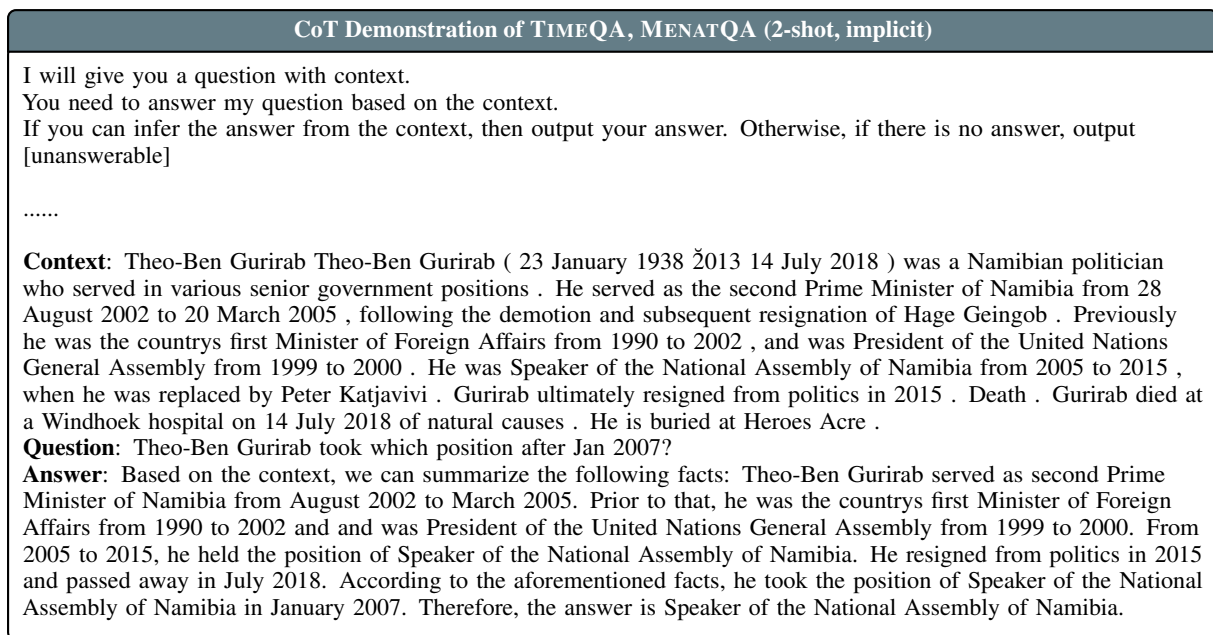


Figure 16: Chain-of-Thought demonstrations of TimeQA, MenatQA, implicit reasoning.

CoT Demonstration of TEMPReason (4-shot, event-time)
I will give you a question with context. You need to answer my question based on the context. Context (facts): Gian Piero Gasperini is the head coach of Atalanta B.C. from Jun, 2016 to Dec, 2022. Edoardo Reja is the head coach of Atalanta B.C. from Mar, 2015 to Jun, 2016. Stefano Colantuono is the head coach of Atalanta B.C. from Jun, 2010 to Mar, 2015. Bortolo Mutti is the head coach of Atalanta B.C. from Jan, 2010 to Jun, 2010. Emiliano Mondonico is the head coach of Atalanta B.C. from Jul, 1987 to Jun, 1990. Marcello Lippi is the head coach of Atalanta B.C. from Jul, 1992 to Jun, 1993. Angelo Gregucci is the head coach of Atalanta B.C. from Jul, 2009 to Sep, 2009. Luigi Delneri is the head coach of Atalanta B.C. from Jul, 2007 to Jun, 2009. Ottavio Bianchi is the head coach of Atalanta B.C. from Jul, 1981 to Jun, 1983. Antonio Conte is the head coach of Atalanta B.C. from Sep, 2009 to Jan, 2010. Nedo Sonetti is the head coach of Atalanta B.C. from Jul, 1983 to Jun, 1987. Valter Bonacina is the head coach of Atalanta B.C. from Jan, 2010 to Jan, 2010. Question: Who was the head coach of the team Atalanta B.C. in Feb, 2016? Answer: According to the context, Edoardo Reja was the head coach of Atalanta B.C. from Mar, 2015 to Jun, 2016. In Feb 2016, the head coach of the team Atalanta B.C. is Edoardo Reja. Therefore, the answer is Edoardo Reja.

Figure 17: Chain-of-Thought demonstrations of TempReason, event-time reasoning.

CoT Demonstration of TEMPReason (4-shot, event-event)
I will give you a question with context. You need to answer my question based on the context. Context (facts): Nicholas Macpherson holds the position of Member of the House of Lords from Oct, 2016 to Dec, 2022. Nicholas Macpherson holds the position of Principal Private Secretary to the Chancellor of the Exchequer from Jan, 1993 to Jan, 1997. Nicholas Macpherson holds the position of Permanent Secretary to the Treasury from Aug, 2005 to Jan, 2016. Question: Which position did Nicholas Macpherson hold before Member of the House of Lords? Answer: According to the context, Nicholas Macpherson holds the position of Permanent Secretary to the Treasury from Aug, 2005 to Jan, 2016. Afterthat, Nicholas Macpherson holds the position of Member of the House of Lords from Oct, 2016 to Dec, 2022. Nicholas Macpherson hold the position of Permanent Secretary to the Treasury before Member of the House of Lords. Therefore, the answer is Permanent Secretary to the Treasury."

Figure 18: Chain-of-Thought demonstrations of TempReason, event-event reasoning.